



Study of Fluctuations in Undulator Radiation in the IOTA Ring at Fermilab

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Collaboration



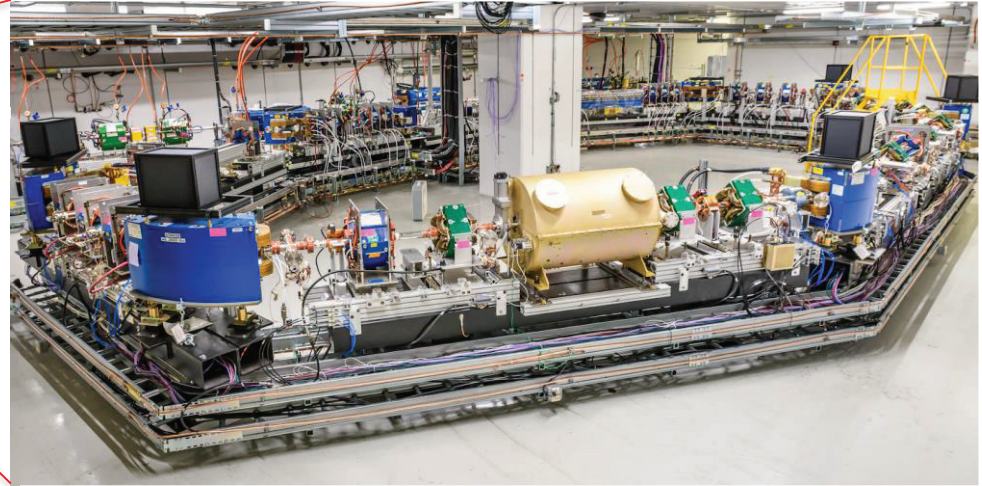
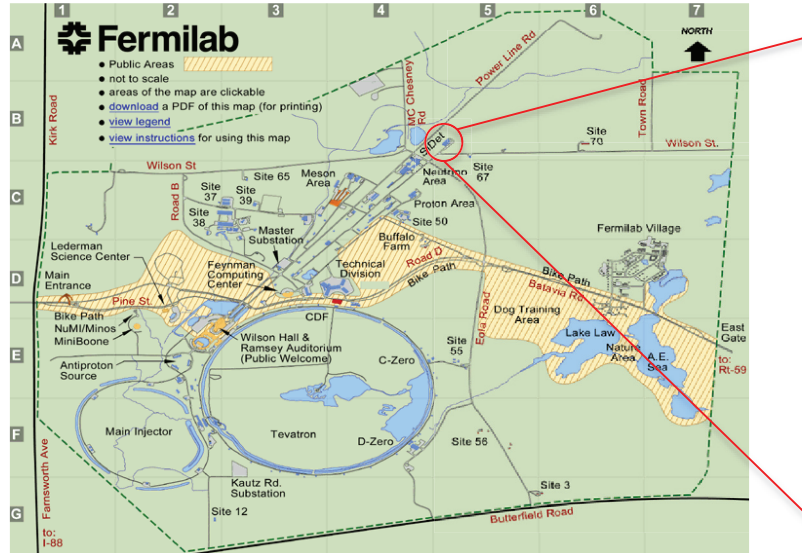
I. Lobach, UChicago, S. Nagaitsev, V. Lebedev, A. Romanov, A. Valishev, G. Stancari, Fermilab, A. Halavanau, Z. Huang, V. Yakimenko, SLAC, A. Murokh, Radiabeam, K. J. Kim, ANL, T. Shaftan, BNL

Acknowledgments:

We would like to thank the entire FAST/IOTA team for helping us with building and installing the setup, and taking measurements, especially Wayne Johnson, Mark Obrycki, and James Santucci; Greg Saewert, for constructing the photodetector circuit and providing the test light source; David Johnson and Todd Johnson, for kindly providing test equipment and assisting during tests of our detector. Numerous pieces of advice given by Daniil Frolov are greatly appreciated as well. A.H. is grateful to G. Stupakov and Y. Cai (SLAC) for many in-depth physics discussions on the subject.

Integrable Optics Test Accelerator (IOTA)

- First beam Aug 21, 2018

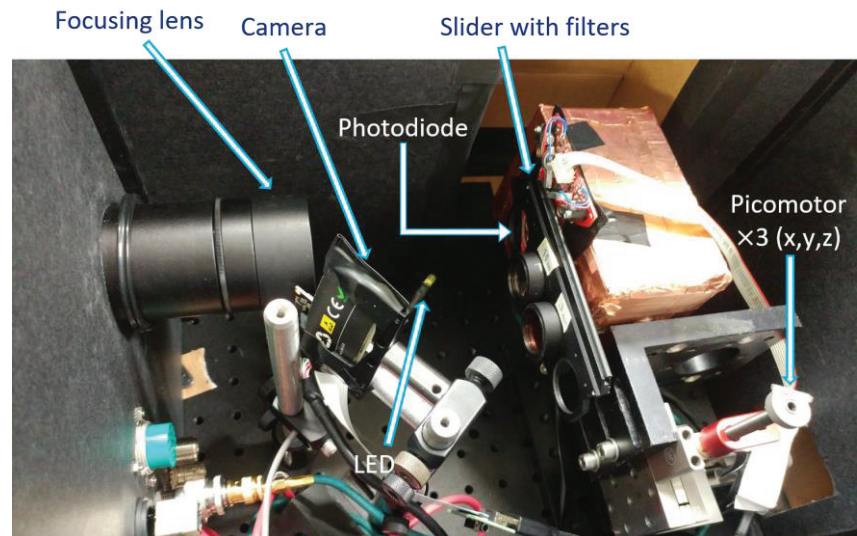


- Particles: electrons/protons
- Main experiments:
 - Nonlinear beam optics
 - Optical stochastic cooling

Circumference: 40 m
Electron energy: 100 MeV

Not a user facility!

Experiment idea

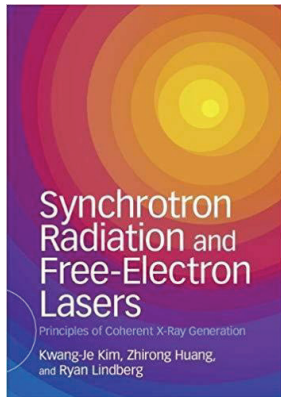


- We installed an undulator in the IOTA ring (late Feb).
- And built an integrating circuit for the photodiode's current. The amplitude of the output voltage was proportional to the number of photoelectrons generated in the photodiode.
- In the experiment (late Mar), we studied the fluctuation in the number of photoelectrons, namely, the variance:

$$\text{var}(\mathcal{N}) = \langle \mathcal{N}^2 \rangle - \langle \mathcal{N} \rangle^2$$

Theoretical prediction

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$$\text{var}(\mathcal{N}_{\text{ph}}) = \langle \mathcal{N}_{\text{ph}} \rangle + \frac{1}{M} \langle \mathcal{N}_{\text{ph}} \rangle^2$$

Discrete quantum
nature of light

Chaotic light. Incoherent
sum over randomly phased
electrons

In our experiment:

#1 Wide band, large solid angle, high QE=80%

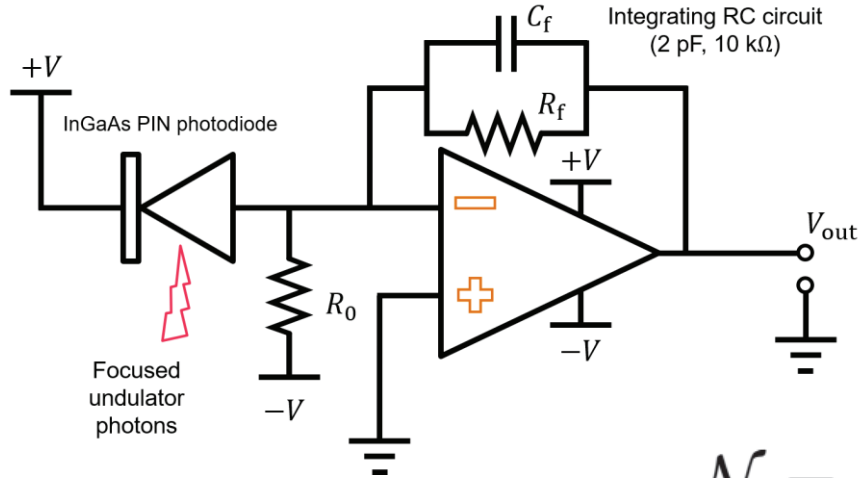
#2 The two terms are comparable

#3 RMS fluctuation $\sim 10^{-4} - 10^{-3}$

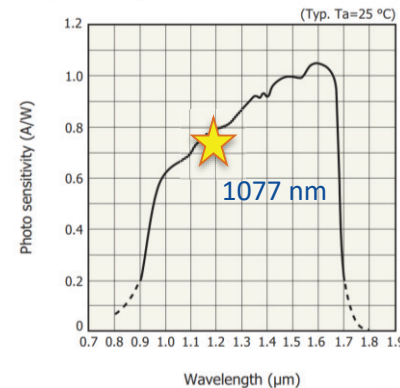
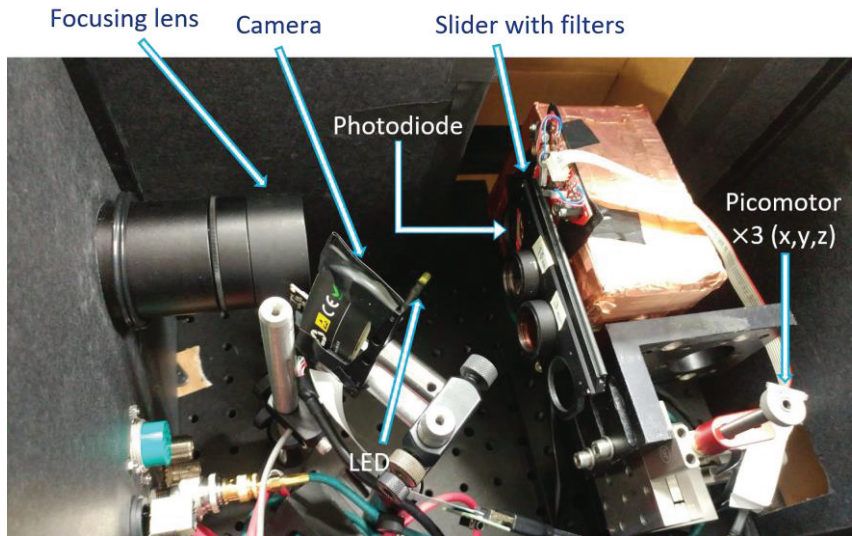
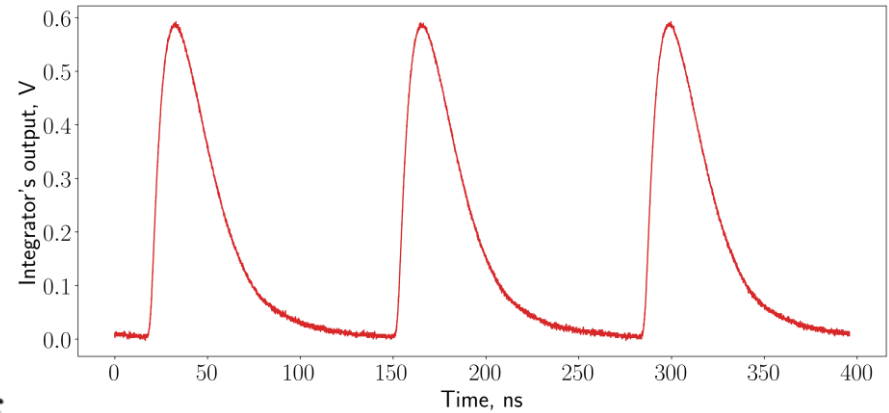
$$\frac{d\mathcal{N}_{\text{q.c.}}^{(1)}}{d\mathbf{k}} = \eta_{\mathbf{k}} I_{\mathbf{k}}^{(1)}$$

$$\frac{1}{M} \equiv \frac{\text{var}(\mathcal{N}_{\text{q.c.}})}{\langle \mathcal{N}_{\text{q.c.}} \rangle^2} = \frac{\frac{\sqrt{\pi}}{\sigma_z} \int dk d\Omega_1 d\Omega_2 k^4 \eta_{k\mathbf{n}_1} I_{k\mathbf{n}_1}^{(1)} \eta_{k\mathbf{n}_2} I_{k\mathbf{n}_2}^{(1)} e^{-k^2 \sigma_x^2 (\theta_{1x} - \theta_{2x})^2 - k^2 \sigma_y^2 (\theta_{1y} - \theta_{2y})^2}}{\left(\int d\mathbf{k} \eta_{\mathbf{k}} I_{\mathbf{k}}^{(1)} \right)^2}$$

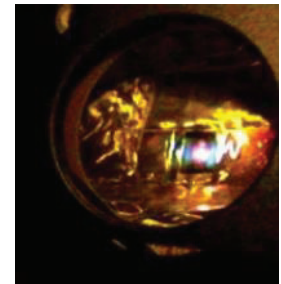
More details on the setup. The photodiode detector



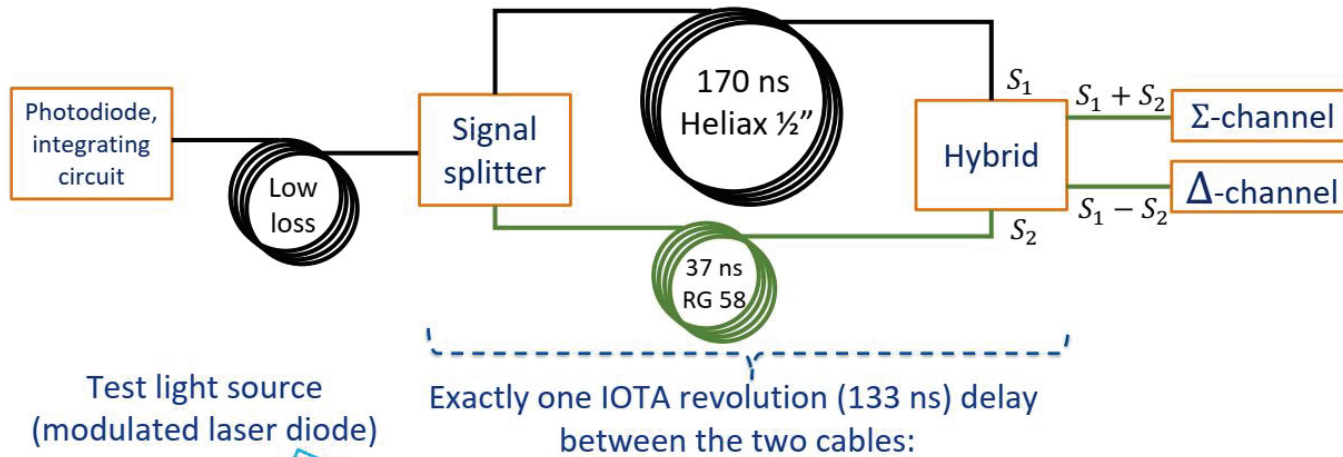
$$\mathcal{N} = \frac{C_f}{e} A$$



*we did not check these data for QE



Comb filter. Testing the setup



$$S_1 = (1 + \delta_1)S(t)$$

$$S_2 = (1 + \delta_2)S(t)$$

$$\Sigma = S_1 + S_2 \approx 2S(t)$$

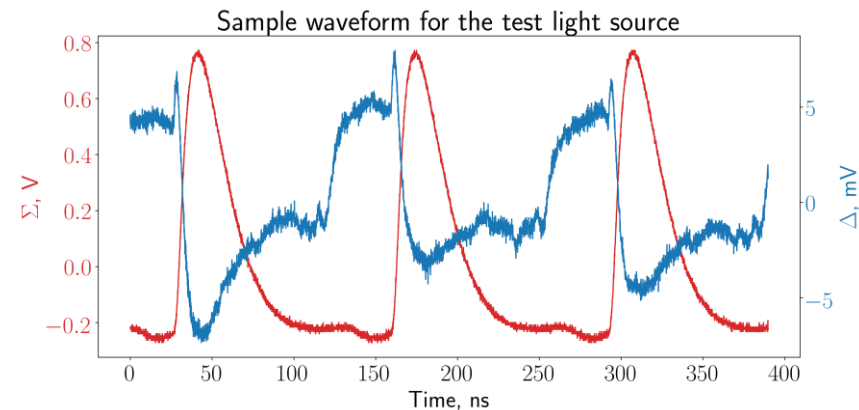
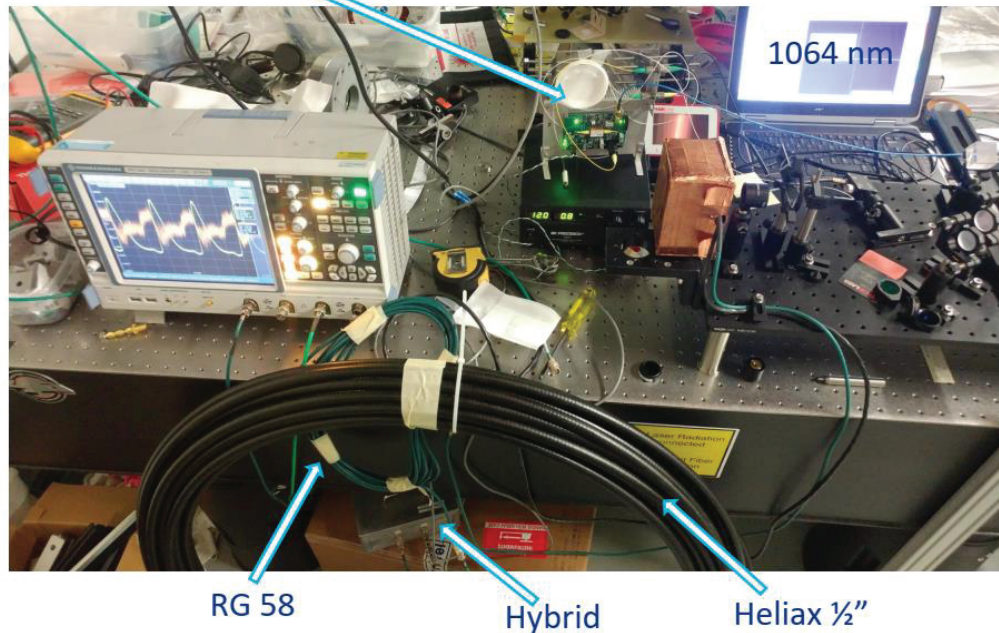
$$\Delta = S_1 - S_2 = (\delta_1 - \delta_2)S(t)$$

$$A = (1 + \delta)S(t_{\max})$$

$$\text{var}(\delta_1 - \delta_2) = 2\text{var}(\delta)$$

$$\text{var}(A) = \text{var}(\Delta(t_{\max}))/2$$

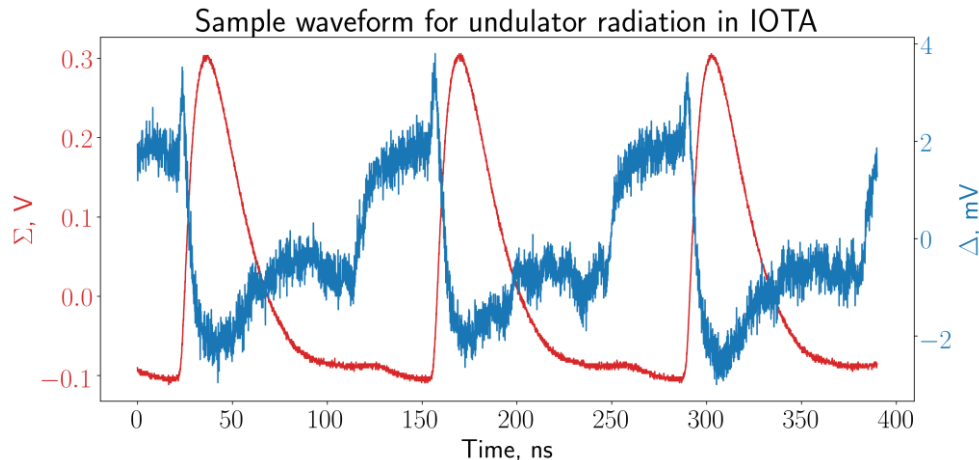
One dataset is 1.5 ms (~11000 turns)



Crosstalk: 0.7%

Noise subtraction algorithm

- In the real experiment with undulator radiation signal/noise < 1:



Main sources of noise:

- The oscilloscope
1mV peak-to-peak
- The integrator's op-amp
1.5mV peak-to-peak
(together with the scope)

Total RMS noise: $\approx 0.3\text{mV}$

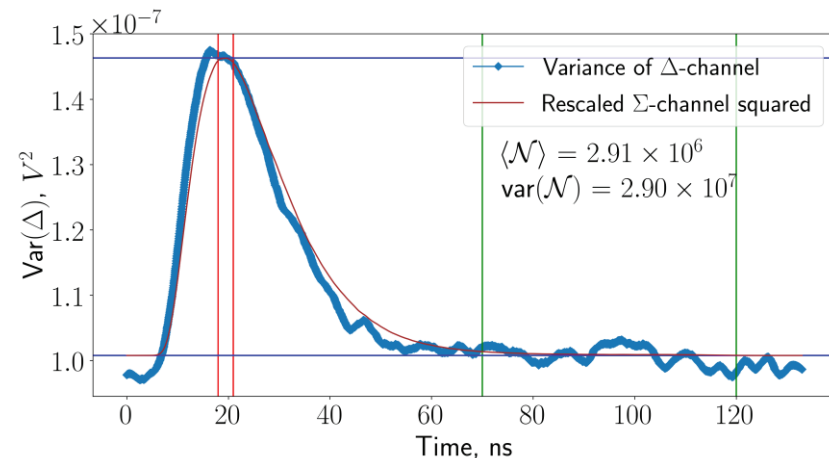
Extraction of sub noise level fluctuations:

- #1 Find the period with high accuracy (>7 figures)
- #2 Map all Δ -channel data to one period
- #3 Bin the data along the time axis
- #4 Take variance of Δ -channel in each bin:

$$\Delta(t) = (\delta_1 - \delta_2)S(t) + \text{noise}$$

@ fixed t $\downarrow$ const

$$\text{var}(\Delta(t)) = 2\text{var}(\delta)S^2(t) + \text{var}(\text{noise})$$



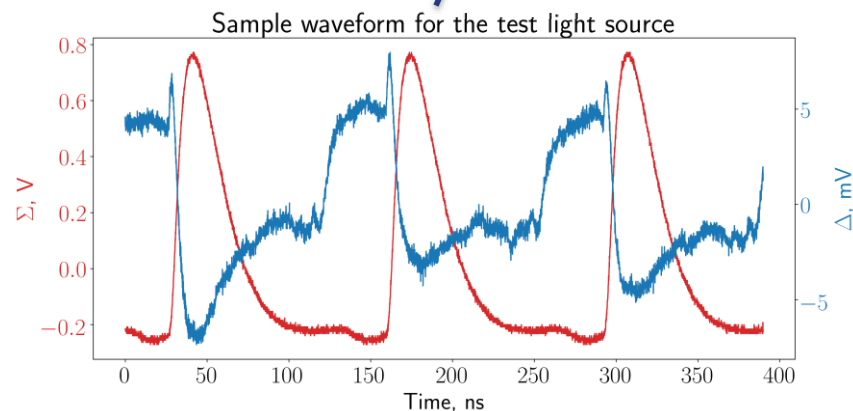
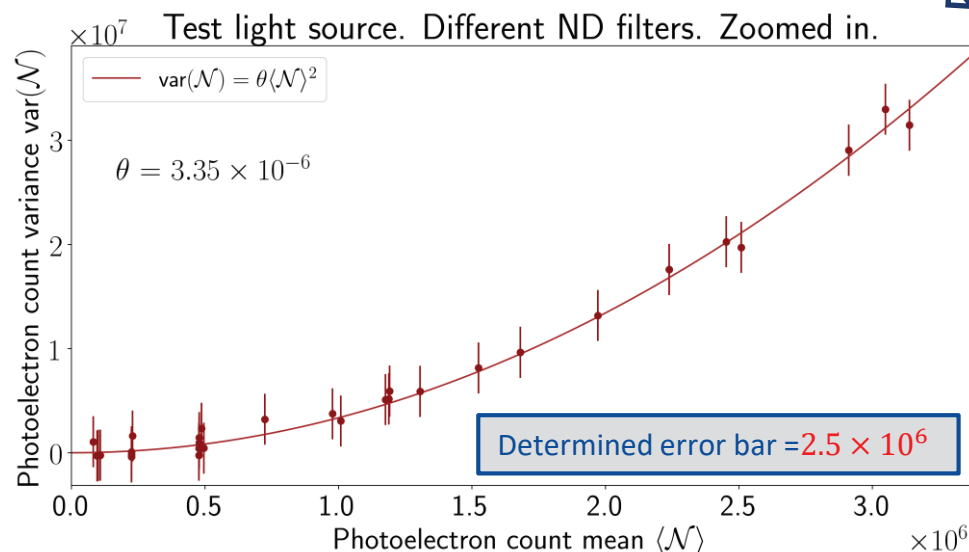
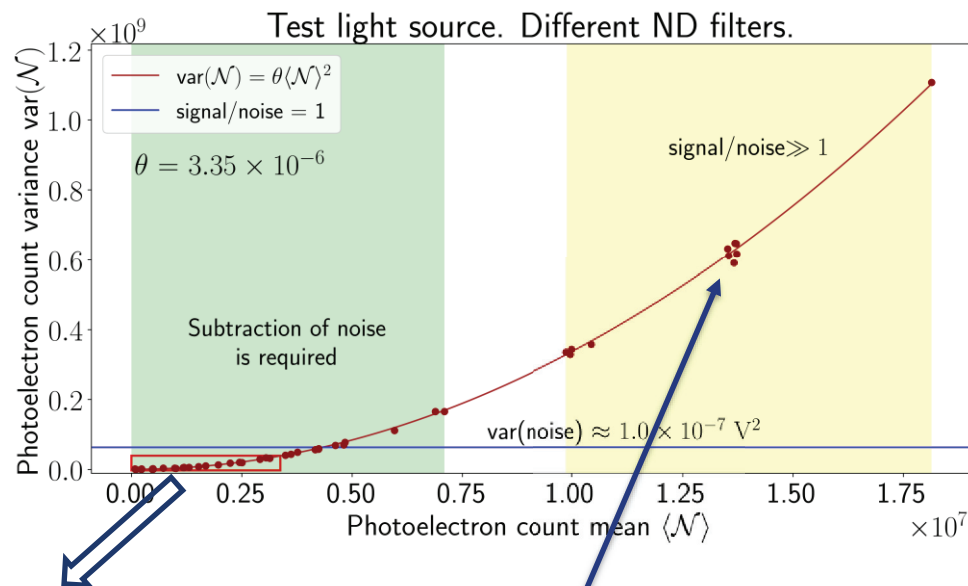
Testing the setup. Finding its precision

Detector test idea:

Keep the test light source in the same regime and use different **ND filters**. Relative classical fluctuation (due to pulse generator errors) must stay the same:

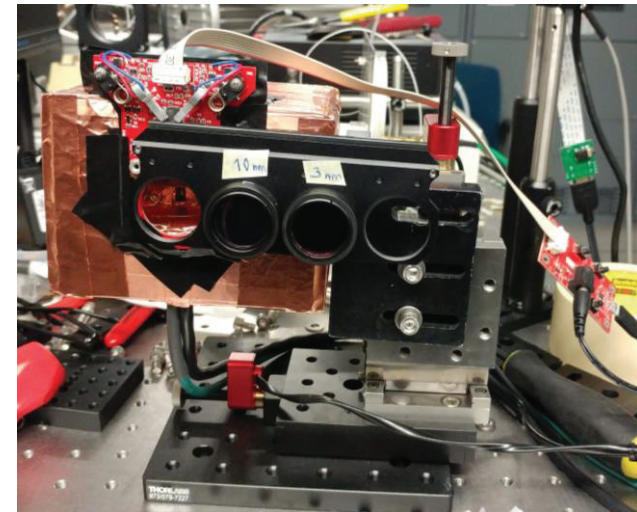
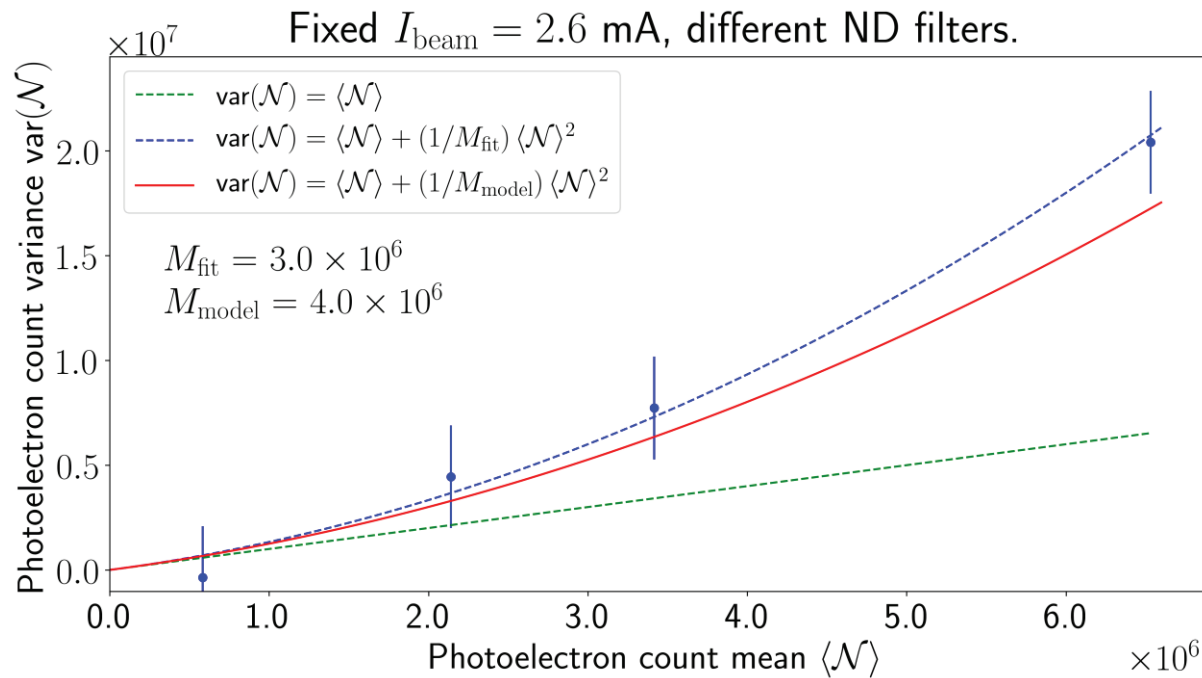
$$\frac{\text{var}(A)}{\langle A \rangle^2} = \frac{\text{var}(\mathcal{N})}{\langle \mathcal{N} \rangle^2} = \theta = \text{const}$$

- θ is determined at large $\langle \mathcal{N} \rangle$, when signal/noise $\gg 1$.
- Poisson contribution is negligible.



Measurement results in IOTA

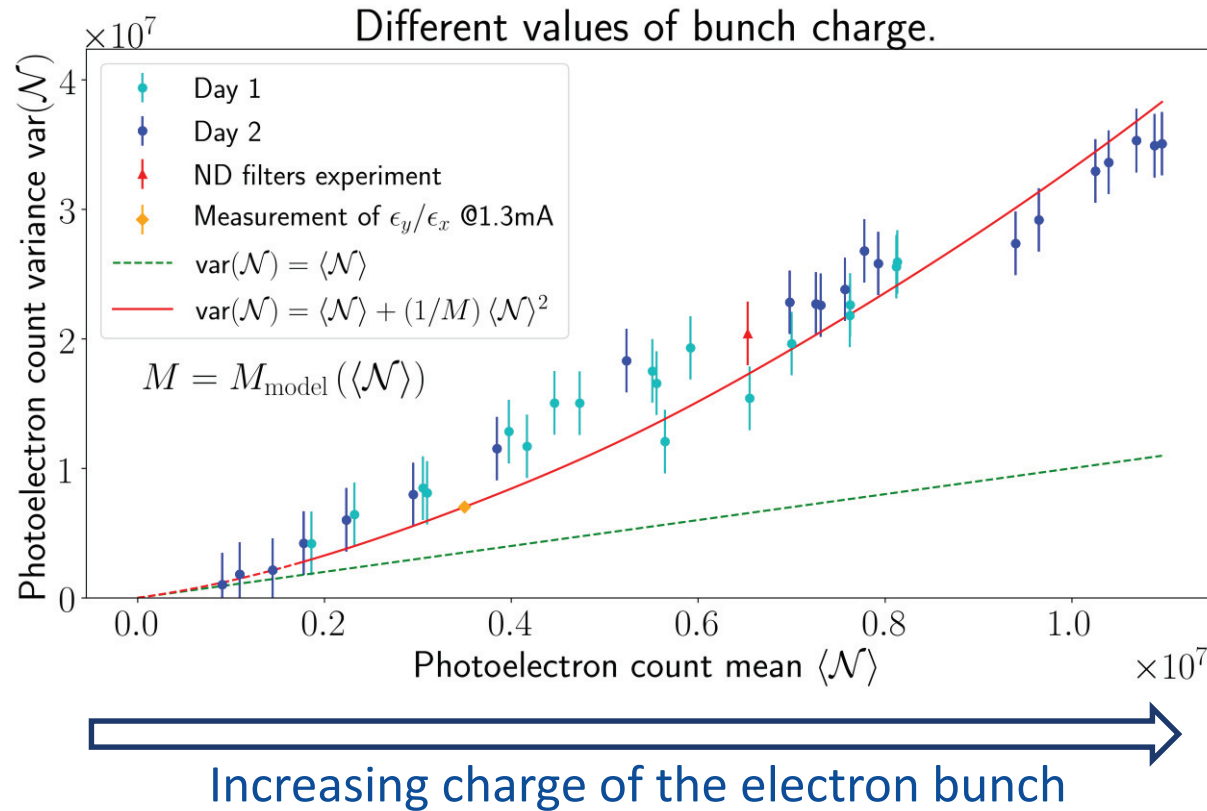
- Constant number of electrons in the bunch, different neutral density filters in front of the detector



Decreasing optical density of the filters

Measurement results in IOTA

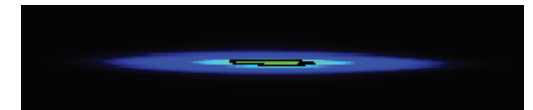
- Changing the electron bunch charge. No neutral density (ND) filters in front of the detector.



Model for bunch dimensions:

- Longitudinal size is constant, measured with a wall-current monitor
- Vertical size is constant and determined by multiple scattering in the background gas
- Horizontal size and momentum spread are determined by **intrabeam scattering***

Measurement of ϵ_y/ϵ_x @1.3mA:



*S. Nagaitsev, *Phys. Rev. ST Accel. Beams* 8.6 (2005): 064403.

Conclusions

- Quantitative theoretical model of the fluctuations in synchrotron radiation was developed and verified in the experiment with undulator radiation in IOTA
- It helped corroborate a model of intrabeam scattering in IOTA. The agreement is expected to improve in the future.
- It was shown that along with measurements of longitudinal bunch size [2-5] the fluctuations can be used to measure transverse bunch size in some cases ,e.g., in IOTA.

I. Lobach, V. Lebedev, S. Nagaitsev, A. Romanov, G. Stancari, A. Halavanau, Z. Huang, and K.-J. Kim, *Intensity fluctuations in undulator radiation*, will be submitted to PRAB.

Improvements as compared to the similar experiment from [1]:

- Better precision due to using the comb filter with one-turn delay and the special noise subtraction algorithm. In IOTA, fluctuations were two orders of magnitude smaller than in [1].
- Fluctuations data were collected for different values of bunch charge.
- The transition from Poisson statistics to Super-Poisson statistics was observed in undulator radiation for the first time.

- [1] M. C. Teich, T. Tanabe, T. C. Marshall, and J. Galayda, Statistical properties of wiggler and bending-magnet radiation from the Brookhaven Vacuum-Ultraviolet electron storage ring, *Phys. Rev. Lett.* **65**, 3393 (1990).
- [2] F. Sannibale, G. Stupakov, M. Zolotorev, D. Filippetto, and L. Jägerhofer, Absolute bunch length measurements by incoherent radiation fluctuation analysis, *Phys. Rev. ST Accel. Beams* **12**, 032801 (2009).
- [3] P. Catravas, W. Leemans, J. Wurtele, M. Zolotorev, M. Babzien, I. Ben-Zvi, Z. Segalov, X.-J. Wang, and

- V. Yakimenko, Measurement of electron-beam bunch length and emittance using shot-noise-driven fluctuations in incoherent radiation, *Phys. Rev. Lett.* **82**, 5261 (1999).
- [4] V. Sajaev, Measurement of bunch length using spectral analysis of incoherent radiation fluctuations, in *AIP Conf. Proc.*, Vol. 732 (AIP, 2004) pp. 73–87.
- [5] V. Sajaev, *Determination of longitudinal bunch profile using spectral fluctuations of incoherent radiation*, Report No ANL/ASD/CP-100935 (Argonne National Laboratory, 2000).



Thank you for your attention!

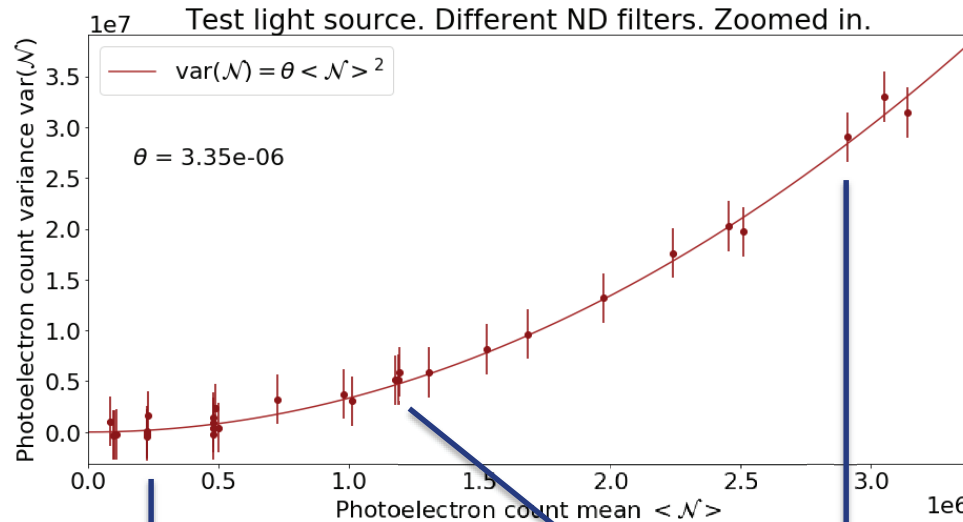


Additional slides

Main experiment parameters

IOTA circumference	40 m (133 ns)
Beam energy	100 MeV
Max average current	4.0 mA
ϵ_x, ϵ_y	0.32 μm , 31 nm
σ_p	3.1×10^{-4}
β_x, β_y	1.82 m, 1.75 m
D_x, D_y	0.87 m, 0 m
σ_x, σ_y	815 μm , 75 μm
σ_z	38 cm
Rad. damping rates $1/\tau_x, 1/\tau_y$	$0.336 \text{ s}^{-1}, 0.852 \text{ s}^{-1}$
$1/\tau_p$	2.22 s^{-1}
Undulator parameter K_u	1.0
Undulator period	55 mm
Number of undulator periods	10
Fundamental harmonic wavelength	1077 nm
Photodiode diameter	1 mm
Quantum efficiency @1077 nm	80 %
Beam lifetime	> 10 min

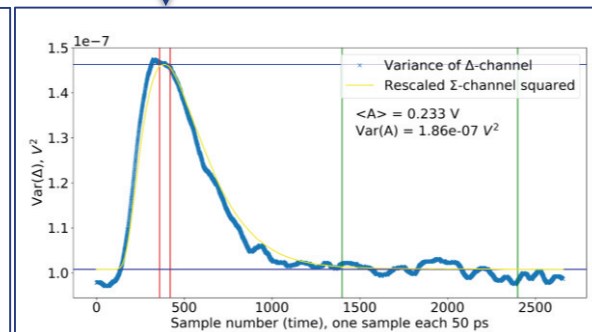
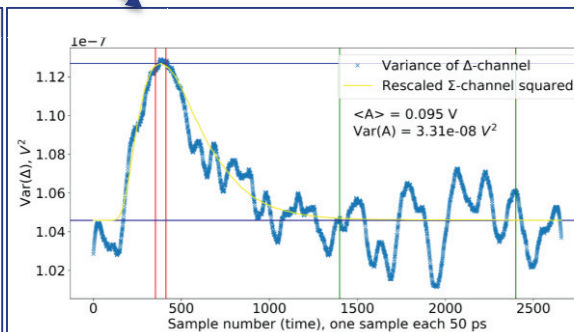
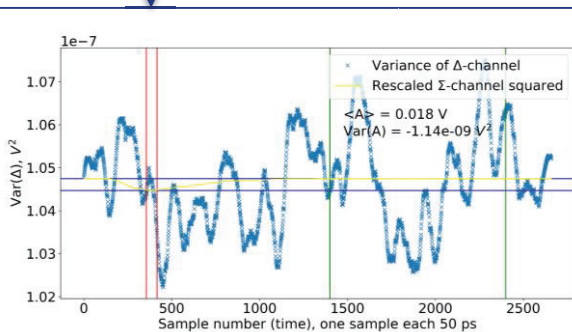
Testing the setup for fluctuation values as in the actual experiment in IOTA



Three standard deviations from the fit give our algorithm's error

Error bar = 2.5×10^6

*in all three plots noise level is the same!



$$\text{var}(\Delta(t)) = 2\text{var}(\delta)S^2(t) + \text{var}(\text{noise}) \quad \text{*in IOTA } \text{var}(\text{noise}) = 0.8 \times 10^{-7} \text{V}^2$$

The experimental data from BNL

